



VOICEINSIGHTS AFRICA

SYNTHETIC DEMONSTRATION PUBLICATION

Energy Access Intelligence Report

Tanzania | Energy Access

3,763 synthetic responses

Prepared by VoiceInsights Africa

Every Voice. Every Language. Every Insight.

Synthetic demonstration. Not official statistics.

Tanzania has made real gains in energy access performance – and real gaps remain alongside them. This publication separates the two, using a governed evidence model built to surface risk and opportunity, not just describe them. No claim in this publication stands alone: findings link to synthetic evidence, recommendations carry a named owner, and every statistic shares one internally consistent dataset.

Sample: 3763 synthetic responses – 86% response rate – 4 regions

Publication quality gate: publication ready



Dodoma Region: 68% performance, 1168 responses, risk WATCH
Mwanza Region: 69% performance, 1016 responses, risk WATCH
Mbeya Region: 70% performance, 865 responses, risk STABLE
Coastal Region: 71% performance, 714 responses, risk STABLE



sex: Women 52% & Men 47% & Other / prefer not to say 1%

age: 18& 24 24% & 25& 39 38% & 40& 59 27% & 60+ 11%

location: Urban 44% & Rural 56%



IND-01 Household electrification rate: 68% vs target 92% â OFF_TRACK
IND-02 Average daily supply-reliability hours: 69% vs target 76% â WATCH
IND-03 Productive-use connection share: 70% vs target 77% â WATCH
IND-04 Rural electrification gap: 71% vs target 78% â WATCH



The aggregate result conceals a material implementation divide. Dodoma Region records a 86% signal for grid-extension versus off-grid economics; the shortfall is concentrated among rural women, where service continuity and confidence move together. This is not a new problem.

Evidence: EVI-ENE-001, EVI-ENE-004 • Confidence 89.5%

54%. That is the supply-reliability gaps signal recorded in Mwanza Region, and the headline figure understates the problem: disaggregated by group, young adults sit well below it. The clearest gap sits in Mwanza Region, concentrated among young adults. A twelve-month delay would let the gap compound into a harder, costlier problem.

Evidence: EVI-ENE-002, EVI-ENE-005 • Confidence 90.5%

What would it take to fix productive-use uptake among newly connected households? Start with Mbeya Region, where the signal sits at 55% and persons with disabilities carry most of the shortfall. The honest answer is sustained attention, not a single intervention.

Mbeya Region shows the sharpest divide, where persons with disabilities carry the largest share of the gap. Left unaddressed for a year, the imbalance is likely to erode equity, delivery confidence and the durability of results.

Evidence: EVI-ENE-003, EVI-ENE-006 • Confidence 91.5%

Progress on connection-to-reliable-supply gap remains visible but not yet durable. The 56% result in Coastal Region weakens after disaggregation, particularly for frontline workers, suggesting that current gains depend on delivery conditions that remain fragile. Of every study area examined in Tanzania, connection-to-reliable-supply gap shows up hardest there, and frontline workers feel it most. A year of inaction would turn a gap into a pattern that is far harder to reverse.

Evidence: EVI-ENE-004, EVI-ENE-007 • Confidence 92.5%

Two things are true at once about rural electrification prioritisation. The national picture is improving, and Dodoma Region – where low-income households report a 57% signal – is not improving fast enough to close the gap. Energy access performance in Tanzania looks stable in aggregate but splits sharply once rural electrification prioritisation is disaggregated. Dodoma Region anchors the low end, and low-income households bear most of that shortfall – a pattern that risks compounding without intervention within the year.

Evidence: EVI-ENE-005, EVI-ENE-008 – Confidence 93.5%

Dodoma Region â Elsewhere, grid-extension versus off-grid economics might be a footnote. For rural women in Dodoma Region, it is the whole story.â

Mwanza Region â Young adults in Mwanza Region frame it simply: when supply-reliability gaps breaks down, everything else does too.â

Mbeya Region â In Mbeya Region, productive-use uptake among newly connected households shapes whether persons with disabilities can turn services into lasting outcomes.â

Coastal Region â Frontline workers in Coastal Region put it plainly: connection-to-reliable-supply gap is what decides whether progress holds or slips back.â

Dodoma Region â Why does Dodoma Region lag behind? Ask low-income households, and the answer comes back to rural electrification prioritisation.â

Mwanza Region â Elsewhere, grid-extension versus off-grid economics might be a footnote. For service managers in Mwanza Region, it is the whole story.â

Mbeya Region â Community leaders in Mbeya Region frame it simply: when supply-reliability gaps breaks down, everything else does too.â

Coastal Region â In Coastal Region, productive-use uptake among newly connected households shapes whether private-sector partners can turn services into lasting outcomes.â

Dodoma Region â Rural women in Dodoma Region put it plainly: connection-to-reliable-supply gap is what decides whether progress holds or slips back.â

Mwanza Region â Why does Mwanza Region lag behind? Ask young adults, and the answer comes back to rural electrification prioritisation.â

Risks and alternative trajectories are modelled to show how VoiceInsights translates evidence into prioritised management choices.



DEC-ENE-01 MEDIUM: Direct the next investment cycle to the higher-return grid-extension versus off-grid split. Owner: Country Director / Programme Executive; Timeline: 6â 18 months; Indicator: Proportion of agreed implementation steps verified on schedule for DEC-ENE-01

DEC-ENE-02 CRITICAL: Address the supply-reliability gap undermining connection value. Owner: Country Director / Programme Executive; Timeline: 0â 90 days; Indicator: Quarterly-verified progress against the milestones agreed for this critical-tier decision (DEC-ENE-02)

DEC-ENE-03 CRITICAL: Support productive-use uptake among newly connected households. Owner: Country Director / Programme Executive; Timeline: 3â 12 months; Indicator: Number of agreed milestones for DEC-ENE-03 confirmed complete at each quarterly review

DEC-ENE-04 HIGH: Prioritise rural electrification in the least-served districts. Owner: Country Director / Programme Executive; Timeline: 3â 12 months; Indicator: Percentage of agreed milestones completed for DEC-ENE-04

DEC-ENE-05 HIGH: Introduce reliability service-level monitoring for connected areas. Owner: Country Director / Programme Executive; Timeline: 3â 12 months; Indicator: Share of the high-tier delivery plan confirmed complete each quarter (DEC-ENE-05)



Direct the next investment cycle to the higher-return grid-extension versus off-grid split.

Strategic priority: MEDIUM

Evidence used: EVI-ENE-001, EVI-ENE-004

Rationale: The aggregate result conceals a material implementation divide. Dodoma Region records a 86% signal for grid-extension versus off-grid economics; the shortfall is concentrated among rural women, where service continuity and confidence move together. This is not a new problem.

Expected benefit: The evidence points toward improved energy access outcomes, though the size of the gain will depend on how consistently the action is resourced.

Expected risk: Likely â though not certain â erosion of momentum if the initial sponsor changes role

Dependencies: Timely data collection for the next reporting cycle; Confirmed budget release; Named executive sponsor

Budget requirement: Low to medium â validate through the formal planning cycle

Owner: Country Director / Programme Executive

Supporting organization: M&E, Finance, Safeguarding and Delivery Units

Timeline: 6â 18 months

Monitoring indicator: Proportion of agreed implementation steps verified on schedule for DEC-ENE-01

Success criteria: At least 80% of milestones achieved with verified evidence

Management response: PENDING FORMAL MANAGEMENT RESPONSE

Follow-up: Assign accountable owner; Validate budget; Approve milestone plan; Review evidence quarterly

Address the supply-reliability gap undermining connection value.

Strategic priority: CRITICAL

Evidence used: EVI-ENE-002, EVI-ENE-005

Rationale: 54%. That is the supply-reliability gaps signal recorded in Mwanza Region, and the headline figure understates the problem: disaggregated by group, young adults sit well below it. The clearest gap sits in Mwanza Region, concentrated among young adults. A twelve-month delay would let the gap compound into a harder, costlier problem.

Expected benefit: A more consistent energy access experience for the groups currently furthest behind.

Expected risk: Weak cross-team coordination during early implementation

Dependencies: Sign-off from the relevant oversight body; Cross-team delivery coordination; Validated operational baseline

Budget requirement: High â€” full business case and multi-year funding commitment required

Owner: Country Director / Programme Executive

Supporting organization: M&E, Finance, Safeguarding and Delivery Units

Timeline: 0â€”90 days

Monitoring indicator: Quarterly-verified progress against the milestones agreed for this critical-tier decision (DEC-ENE-02)

Success criteria: A majority of milestones achieved, each independently verified against real evidence

Management response: PENDING FORMAL MANAGEMENT RESPONSE

Follow-up: Assign accountable owner; Validate budget; Approve milestone plan; Review evidence quarterly

Support productive-use uptake among newly connected households.

Strategic priority: CRITICAL

Evidence used: EVI-ENE-003, EVI-ENE-006

Rationale: What would it take to fix productive-use uptake among newly connected households? Start with Mbeya Region, where the signal sits at 55% and persons with disabilities carry most of the shortfall. The honest answer is sustained attention, not a single intervention. Mbeya Region shows the sharpest divide, where persons with disabilities carry the largest share of the gap. Left unaddressed for a year, the imbalance is likely to erode equity, delivery confidence and the durability of results.

Expected benefit: Faster, better-evidenced energy access decisions that donor leadership can act on with confidence.

Expected risk: Competing priorities crowding out sustained follow-through

Dependencies: Named executive sponsor; Sign-off from the relevant oversight body; Timely data collection for the next reporting cycle

Budget requirement: Medium â detailed costing and fiduciary review required

Owner: Country Director / Programme Executive

Supporting organization: M&E, Finance, Safeguarding and Delivery Units

Timeline: 3â 12 months

Monitoring indicator: Number of agreed milestones for DEC-ENE-03 confirmed complete at each quarterly review

Success criteria: Milestone completion confirmed by evidence review, not self-reported progress

Management response: PENDING FORMAL MANAGEMENT RESPONSE

Follow-up: Assign accountable owner; Validate budget; Approve milestone plan; Review evidence quarterly

Prioritise rural electrification in the least-served districts.

Strategic priority: HIGH

Evidence used: EVI-ENE-004, EVI-ENE-007

Rationale: Progress on connection-to-reliable-supply gap remains visible but not yet durable. The 56% result in Coastal Region weakens after disaggregation, particularly for frontline workers, suggesting that current gains depend on delivery conditions that remain fragile. Of every study area examined in Tanzania, connection-to-reliable-supply gap shows up hardest there, and frontline workers feel it most. A year of inaction would turn a gap into a pattern that is far harder to reverse.

Expected benefit: Reduced variation in energy access delivery and a stronger evidence base for the next planning cycle.

Expected risk: Fragmented accountability once implementation moves beyond the pilot stage

Dependencies: Validated operational baseline; Named executive sponsor; Sign-off from the relevant oversight body

Budget requirement: Medium – likely fundable from existing lines, pending confirmation

Owner: Country Director / Programme Executive

Supporting organization: M&E, Finance, Safeguarding and Delivery Units

Timeline: 3 – 12 months

Monitoring indicator: Percentage of agreed milestones completed for DEC-ENE-04

Success criteria: Verified evidence of at least 80% delivery against the agreed milestone plan

Management response: PENDING FORMAL MANAGEMENT RESPONSE

Follow-up: Assign accountable owner; Validate budget; Approve milestone plan; Review evidence quarterly

Introduce reliability service-level monitoring for connected areas.

Strategic priority: HIGH

Evidence used: EVI-ENE-005, EVI-ENE-008

Rationale: Two things are true at once about rural electrification prioritisation. The national picture is improving, and Dodoma Region – where low-income households report a 57% signal – is not improving fast enough to close the gap. Energy access performance in Tanzania looks stable in aggregate but splits sharply once rural electrification prioritisation is disaggregated. Dodoma Region anchors the low end, and low-income households bear most of that shortfall – a pattern that risks compounding without intervention within the year.

Expected benefit: A stronger, more defensible case for energy access investment in the next planning cycle.

Expected risk: Execution capacity and coordination risk

Dependencies: Quarterly evidence review; Validated operational baseline; Named executive sponsor

Budget requirement: Low to medium – validate through the formal planning cycle

Owner: Country Director / Programme Executive

Supporting organization: M&E, Finance, Safeguarding and Delivery Units

Timeline: 3–12 months

Monitoring indicator: Share of the high-tier delivery plan confirmed complete each quarter (DEC-ENE-05)

Success criteria: At least 80% of milestones achieved with verified evidence

Management response: PENDING FORMAL MANAGEMENT RESPONSE

Follow-up: Assign accountable owner; Validate budget; Approve milestone plan; Review evidence quarterly

research objectives: Assess energy access performance and equity; Identify operational drivers and decision priorities

evaluation questions: What outcomes are changing?; Who is being left behind?; Which actions are most likely to improve performance?

sampling frame: Synthetic programme population frame for Tanzania

sample size: 3763

stratification: Region; Location type; Sex; Age; Disability

weights: Normalized post-stratification weights

confidence intervals: 95%

design effect: 1.6

missing data: Documented and sensitivity-tested

reliability: Cronbach alpha 0.86

validity: Content and construct validity checks

metadata: Complete synthetic data dictionary and lineage registry

Publication level: 34 generated publication pages

protocol: Governed synthetic research protocol with sampling, weighting, reliability, validity and reproducibility controls

ethics: Synthetic records only; no real participants or personal data

peer review: Independent synthetic-publication review checklist completed

reproducibility: Deterministic inputs, versioned dataset and export manifest

inferential statistics: Enabled where assumptions and design permit

segmentation: Evidence-led segments

trend analysis: Three-period synthetic series

DEC-ENE-01: Direct the next investment cycle to the higher-return grid-extension versus off-grid split. Owner Country Director / Programme Executive 6 18 months Low to medium validate through the formal planning cycle cost Indicator Proportion of agreed implementation steps verified on schedule for DEC-ENE-01

DEC-ENE-02: Address the supply-reliability gap undermining connection value. Owner Country Director / Programme Executive 0 90 days High full business case and multi-year funding commitment required cost Indicator Quarterly-verified progress against the milestones agreed for this critical-tier decision (DEC-ENE-02)

DEC-ENE-03: Support productive-use uptake among newly connected households. Owner Country Director / Programme Executive 3 12 months Medium detailed costing and fiduciary review required cost Indicator Number of agreed milestones for DEC-ENE-03 confirmed complete at each quarterly review

DEC-ENE-04: Prioritise rural electrification in the least-served districts. Owner Country Director / Programme Executive 3 12 months Medium likely fundable from existing lines, pending confirmation cost Indicator Percentage of agreed milestones completed for DEC-ENE-04

DEC-ENE-05: Introduce reliability service-level monitoring for connected areas. Owner Country Director / Programme Executive 3 12 months Low to medium validate through the formal planning cycle cost Indicator Share of the high-tier delivery plan confirmed complete each quarter (DEC-ENE-05)

EVI-ENE-001: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 88%
EVI-ENE-002: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 89%
EVI-ENE-003: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 90%
EVI-ENE-004: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 91%
EVI-ENE-005: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 92%
EVI-ENE-006: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 93%
EVI-ENE-007: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 94%
EVI-ENE-008: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 95%
EVI-ENE-009: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 96%
EVI-ENE-010: [object Object] â€ APPROVED_SYNTHETIC_DEMONSTRATION â€ confidence 97%

Executive Intelligence Book (2â 7): Executive brief; Decision snapshot; Risks; Opportunities; Budget implications

Evidence Intelligence Book (8â 15): Critical findings; Respondent voice; Evidence chains; Citation registry

Statistical Intelligence Book (16â 22): Sampling; Weighting; Uncertainty; Reliability; Model diagnostics

Decision Intelligence Book (23â 27): Decision matrix; Ownership; Budget; Roadmap; Management response

Standards & Assurance (28â 31): Relevant frameworks; Safeguarding; Privacy; Responsible AI; Accessibility

Technical Appendices (32â 34): Data dictionary; Quality gate; Export manifest; Responsible-use notice

SDG 7: APPLIED_WHERE_RELEVANT â Mapped to EVI-ENE-001, EVI-ENE-002; Synthetic demonstration mapping; institutional validation not implied

Climate Resilience: APPLIED_WHERE_RELEVANT â Mapped to EVI-ENE-002, EVI-ENE-003; Synthetic demonstration mapping; institutional validation not implied

Data Protection: APPLIED_WHERE_RELEVANT â Mapped to EVI-ENE-003, EVI-ENE-004; Synthetic demonstration mapping; institutional validation not implied

Evidence Traceability: APPLIED â Control linked to EVI-ENE-001 and the deterministic publication gate.

Research Ethics: APPLIED â Control linked to EVI-ENE-002 and the deterministic publication gate.

Data Protection: APPLIED â Control linked to EVI-ENE-003 and the deterministic publication gate.

Accessibility: APPLIED â Control linked to EVI-ENE-004 and the deterministic publication gate.

Responsible AI: APPLIED â Control linked to EVI-ENE-005 and the deterministic publication gate.

Publication Integrity: APPLIED â Control linked to EVI-ENE-006 and the deterministic publication gate.

SDG 7: undefined contribution â indicators â undefined

SDG 7: undefined contribution â indicators â undefined

SDG 5: undefined contribution â indicators â undefined

Relevance: Synthetic evidence indicates a strong relevance signal, subject to the stated design limitations. (75/100). Review relevance evidence at the next formal management gate.

Coherence: Synthetic evidence indicates a moderate-to-strong coherence signal, subject to the stated design limitations. (76/100). Review coherence evidence at the next formal management gate.

Effectiveness: Synthetic evidence indicates a strong effectiveness signal, subject to the stated design limitations. (77/100). Review effectiveness evidence at the next formal management gate.

Efficiency: Synthetic evidence indicates a moderate-to-strong efficiency signal, subject to the stated design limitations. (78/100). Review efficiency evidence at the next formal management gate.

Impact: Synthetic evidence indicates a strong impact signal, subject to the stated design limitations. (79/100). Review impact evidence at the next formal management gate.

Sustainability: Synthetic evidence indicates a moderate-to-strong sustainability signal, subject to the stated design limitations. (80/100). Review sustainability evidence at the next formal management gate.

Impact: Sustained, equitable improvement in energy access performance

OC-01: Improved energy access delivery and accountability â indicators IND-01, IND-02

OC-02: More equitable and durable outcomes â indicators IND-03, IND-04

OUT-01: Verified findings and interpreted decision products

OUT-02: Assigned actions with monitoring indicators

Assumptions: Leadership ownership; Safeguarding and privacy controls remain effective

Means of verification: Evidence registry; Quarterly management review; Follow-up measurement

Commitment 1: CHS Commitment 1 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 2: CHS Commitment 2 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 3: CHS Commitment 3 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 4: CHS Commitment 4 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 5: CHS Commitment 5 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 6: CHS Commitment 6 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 7: CHS Commitment 7 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 8: CHS Commitment 8 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

Commitment 9: CHS Commitment 9 â CONTEXT_ONLY. Action: Validate relevance and assign an accountable owner before live use.

EVI-ENE-001 â€ survey â€ Dodoma Region â€ confidence 88% â€ APPROVED
EVI-ENE-002 â€ key_informant_interview â€ Mwanza Region â€ confidence 89% â€ APPROVED
EVI-ENE-003 â€ administrative_record â€ Mbeya Region â€ confidence 90% â€ APPROVED
EVI-ENE-004 â€ survey â€ Coastal Region â€ confidence 91% â€ APPROVED
EVI-ENE-005 â€ key_informant_interview â€ Dodoma Region â€ confidence 92% â€ APPROVED
EVI-ENE-006 â€ administrative_record â€ Mwanza Region â€ confidence 93% â€ APPROVED
EVI-ENE-007 â€ survey â€ Mbeya Region â€ confidence 94% â€ APPROVED
EVI-ENE-008 â€ key_informant_interview â€ Coastal Region â€ confidence 95% â€ APPROVED
EVI-ENE-009 â€ administrative_record â€ Dodoma Region â€ confidence 96% â€ APPROVED
EVI-ENE-010 â€ survey â€ Mwanza Region â€ confidence 97% â€ APPROVED

indicator_1: Synthetic Energy Access indicator 1; numeric; allowed: 0â 100

indicator_2: Synthetic Energy Access indicator 2; categorical; allowed: Documented code list

indicator_3: Synthetic Energy Access indicator 3; categorical; allowed: Documented code list

indicator_4: Synthetic Energy Access indicator 4; numeric; allowed: 0â 100

indicator_5: Synthetic Energy Access indicator 5; categorical; allowed: Documented code list

indicator_6: Synthetic Energy Access indicator 6; categorical; allowed: Documented code list

indicator_7: Synthetic Energy Access indicator 7; numeric; allowed: 0â 100

indicator_8: Synthetic Energy Access indicator 8; categorical; allowed: Documented code list

indicator_9: Synthetic Energy Access indicator 9; categorical; allowed: Documented code list

indicator_10: Synthetic Energy Access indicator 10; numeric; allowed: 0â 100

indicator_11: Synthetic Energy Access indicator 11; categorical; allowed: Documented code list

indicator_12: Synthetic Energy Access indicator 12; categorical; allowed: Documented code list

Synthetic Demonstration Publication produced by VoiceInsights Africa. All people, quotations, locations and statistics are synthetic, internally governed demonstration content and are not official statistics.

All content is synthetic and must not be interpreted as official statistics

Geographic references are illustrative

Budget implications are directional and require formal costing

Causal claims are not made without an appropriate design

Status: DEMONSTRATION_READY

Passed: Editorial Review Passed

Passed: Evidence Review Passed

Passed: Publication Integrity Verified

Passed: Accessibility Verified

Passed: Export Validation Passed

Passed: Statistical Review Passed

Open gates: VISUAL_QA, HUMAN_REVIEW

Verification checks cover weighted evidence, statistical, visual, accessibility, export and review rules. Synthetic samples are labelled Demonstration Ready.